

BPC-iMX8MP-03 Industrial Computer User Guide

Version: V1.2(2025-09)

Compiled by: Polyhex Technology Company Limited (<http://www.polyhex.net/>)

BPC-iMX8MP-03 Industrial Computer is a ruggedized and protected computer. It is composed of a DEBIX SOM A (core board), a DEBIX SOM A I/O board (carrier board) and a steel and aluminum enclosure. It combines various types of harsh environment resistance features, including ruggedness, dustproof, anti-vibration, shock resistance, wide temperature, portability and other indicators.



Figure 1

REVISION HISTORY		
Rev.	Date	Description
1.0	2023.03.17	First edition
1.1	2025.03.21	Updated the part of 1.3.technical support and optimized content
1.2	2025.09.18	Updated the Table 2 product specification

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Chapter 1 Security

1.1. Safety Precaution

This document inform how to make each cable connection. In most cases, you will simply need to connect a standard cable.

Table 1 Terms and conventions

Symbol	Meaning
<i>Warning!</i> 	Always disconnect the power cord from the chassis whenever there is no workload required on it. Do not connect the power cable while the power is on. Sudden power surges can damage sensitive electronic components. Only experienced electricians should open the chassis.
<i>Caution!</i> 	Always ground yourself to remove any static electric charge before touching <i>BPC-iMX8MP-03</i> product. Modern electronic devices are very sensitive to electric charges. Use a grounding wrist strap at all times. Place all electronic components on a static-dissipative surface or in a static-shielded bag.

1.2. Safety Instruction

To avoid malfunction or damage to this product please observe the following:

1. Disconnect the device from the DC power supply before cleaning. Use a damp cloth. Do not use liquid detergents or spray-on detergents.
2. Keep the device away from moisture.
3. During installation, set the device down on a reliable surface. Drops and bumps will lead to damage.
4. Before connecting the power supply, ensure that the voltage is in the required range, and the way of wiring is correct.
5. Carefully put the power cable in place to avoid stepping on it.
6. If the device is not used for a long time, power it off to avoid damage caused by sudden

overvoltage.

7. Do not pour liquid into the venting holes of the enclosure, as this could cause fire or electric shock.

8. For safety reasons, the device can only be disassembled by professional personnel.

9. If one of the following situations occur, get the equipment checked by service personnel:

- The power cord or plug is damaged.
- Liquid has penetrated into the equipment.
- The equipment has been exposed to moisture.
- The equipment does not work well, or you cannot get it to work according to the user's manual.
- The equipment has been dropped and damaged.
- The equipment has obvious signs of breakage.

10. Do not place the device in a place where the ambient temperature is below -40°C (-4°F) or above 85°C (158°F). This will damage the machine. It needs to be kept in an environment at controlled temperature.

11. Due to the sensitive nature of the equipment, it must be stored in a restricted access location, only accessible by qualified engineer.

DISCLAIMER: Polyhex assumes no liability for the accuracy of any statement of this instructional document.

1.3. Technical Support

1. Visit DEBIX website <https://www.debix.io/> where you can find the latest information about the product.

■ Quick Links:

Debix Documentation: <https://debix.io/Document/manual.html>

Debix Blog: <https://debix.io/Software/blog.html>

Debix GitHub: <https://github.com/debix-tech>

2. Contact your distributor, sales representative or Polyhex's customer service center for technical support if you need additional assistance. Please have the following info ready before you call:

- Product name and memory size
- Description of your peripheral attachments
- Description of your software(operating system, version, application software, etc.)
- A complete description of the problem
- The exact wording of any error messages

■ TechSupport Platforms:

Discord Community (recommended): <https://discord.com/invite/adaHHaDkH2>

Email: teksupport@debix.io

Chapter 2 BPC-iMX8MP-03 Industrial Computer Introduction

BPC-iMX8MP-03 Industrial Computer is a compact and secure computer, which designed for industrial applications and can be widely used in machine vision and machine learning, advanced multimedia, smart cities, edge computing and other solutions that require high reliability.

Main features:

- Stainless steel and aluminum enclosure with MTBF>50,000 hours
- It encompasses a DEBIX SOM A and DEBIX SOM A I/O Board
- Support boot from eMMC, Micro SD card and SPI NOR Flash
- Support dual Gigabit Ethernet, 2.4GHz & 5GHz dual-band Wi-Fi 5, Bluetooth 5.0 and a Mini PCIe 4G module/LoRa module/network card
- Support 4 x USB 3.0, 2 x USB 2.0, physically isolated interfaces: 4 x RS232, 2 x RS485/RS232/CAN, 4 x DI, 4 x DO
- Support for Ubuntu, Yocto, Windows 10 IoT, Debian, Android, OpenWRT, and FreeRTOS

2.1. Overview

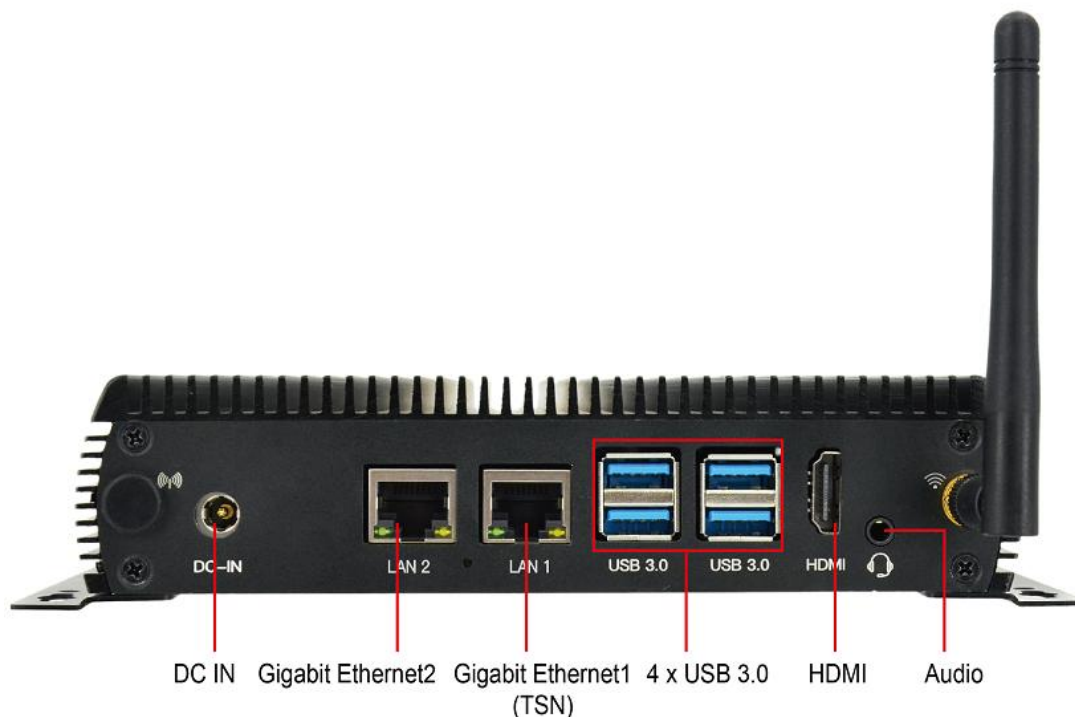


Figure 2

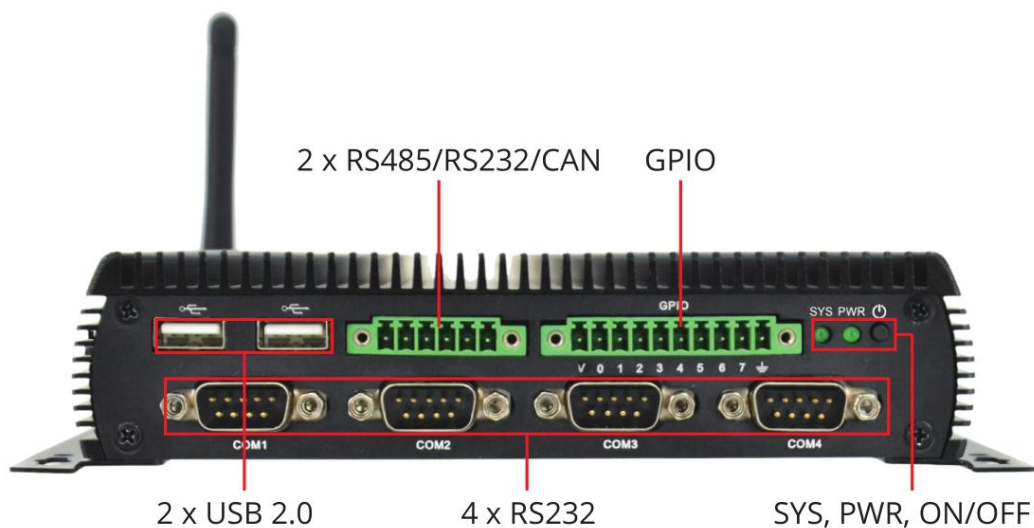


Figure 3

BPC-iMX8MP-03 Industrial Computer uses DEBIX SOM A and DEBIX SOM A I/O Board connection as the main board, which supports dual Gigabit Ethernet, three boot modes, shock and vibration resistance, etc.. The data specifications are as follows.

Table 2 BPC-iMX8MP-03 Industrial Computer specification

BPC-iMX8MP-03	
System	
Motherboard	DEBIX SOM A + DEBIX SOM A I/O Board
Model	BPC-iMX8MP-03
Memory	2GB LPDDR4 (1GB/4GB/8GB optional)
Storage	Onboard 16GB eMMC (8GB/32GB/64GB/128GB/256GB optional)
OS	Ubuntu 22.04, Yocto-L6.1.22_2.2.0, Windows 10 IoT Enterprise, Debian 12 (also supports Android, OpenWRT and FreeRTOS.)
Boot Mode	1) Support DEBIX SOM A board eMMC boot (default) 2) Support DEBIX SOM A I/O Board Micro SD card boot 3) Support DEBIX SOM A I/O Board SPI Nor Flash boot (reserved)
Communication	
Gigabit Ethernet	2 x Independent MAC RJ45 Gigabit Ethernet ports, both support POE power supply (need POE power device module) (One supports TSN)
Wi-Fi & Bluetooth	2.4GHz&5GHz dual-band Wi-Fi 5, Bluetooth 5.0, external Wi-Fi and 4G SMA antenna interface
Video & Audio	
HDMI	1 x HDMI output, connector is Type A HDMI female
Audio	1 x headphone output and microphone input combo interface, the connector is a 3.5mm socket
External I/O Interface	
DC Block	1 x DC socket, supports 5.5mm x 2.1mm plug
USB 3.0	4 x USB 3.0 Host, the connector is double layer Type-A interface
USB 2.0	3 x USB 2.0 Host, the connector is Type-A interface
Serial Ports	1) 4 x physically isolated RS232, compatible with UART TTL 3.3V

	without physical isolation 2) 2 x physically isolated RS485/RS232/CAN (RS485 by default), compatible with UART TTL 3.3V without physical isolation
GPIO	1) 4 x physically isolated DIs, supporting wet and dry nodes 2) 4 x physically isolated DOs, support wet nodes, compatible with external relay dry nodes
LED & Key	1) 1 x System status indicator 2) 1 x Power status indicator 3) 1 x ON/OFF key
Internal I/O Interface	
Slot	1) 1 x Micro SIM pop-up card slot 2) 1 x Micro SD pop-up card slot
Mini PCIe	1) Support Mini PCIe 4G module, such as Quectel 4G module, built-in SIM card 2) Support Mini PCIe LoRa module 3) Support Mini PCIe expansion modules, such as network card, SATA card, serial port card
Power Supply	
Power Input	Default DC 12V/2A power input, support DC 12V~36V wide voltage input
Mechanical & Environmental	
Enclosure Material	Steel and aluminum alloy
Dimension (W x D x H)	124mm x 169.42mm x 38.9mm
Gross Weight	812g
Heat Dissipation	No fan, heat dissipation through the enclosure
Operating Temp.	-20 °C to 70 °C (-40 °C to 85 °C optional)

Relative Humidity	10%~90%
-------------------	---------

2.2. Composition



Figure 4

BPC-iMX8MP-03 Industrial Computer assembly consists of these main components: DEBIX SOM A + I/O Board, enclosure and antenna.

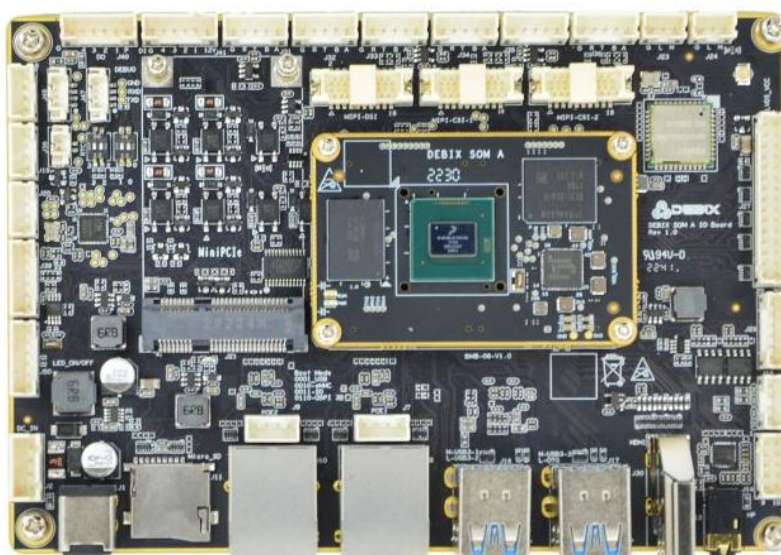


Figure 5 DEBIX SOM A + I/O Board



Figure 6 Enclosure and antenna

2.3. External Interface

2.3.1. Power Interface

BPC-iMX8MP-03 Industrial Computer provides one power connector (DC socket), with default DC 12V/2A power input. As shown in the figure below.



Figure 7 DC-IN

2.3.2. HDMI Interface

BPC-iMX8MP-03 Industrial Computer provides an HDMI interface, and the connector is an A-type HDMI female socket, which is used to connect a monitor, TV or projector. As shown in the Figure 8.

HDMI resolution up to 3840x2160p30. Audio supports 32 channel audio output and supports 1 S/PDIF audio eARC input.

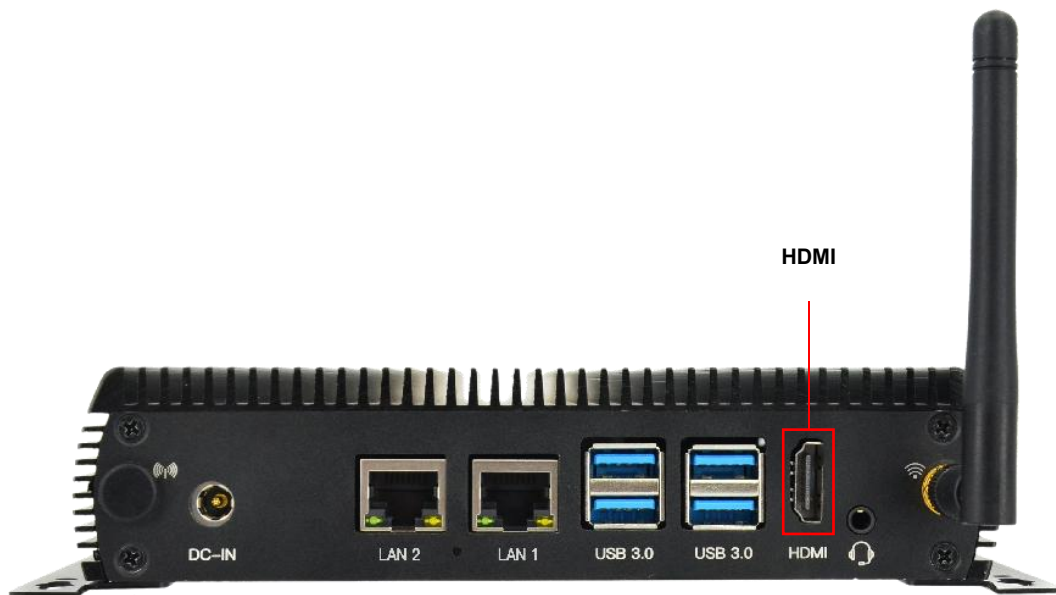


Figure 8 HDMI

2.3.3. Audio Interface

BPC-iMX8MP-03 Industrial Computer provides a combined headphone and microphone input interface. The connector is a 3.5mm socket, compatible with the built-in needle socket design, has audio input/output functions, and supports rated voltage 1.5V MIC audio input. As shown in the Figure 9.

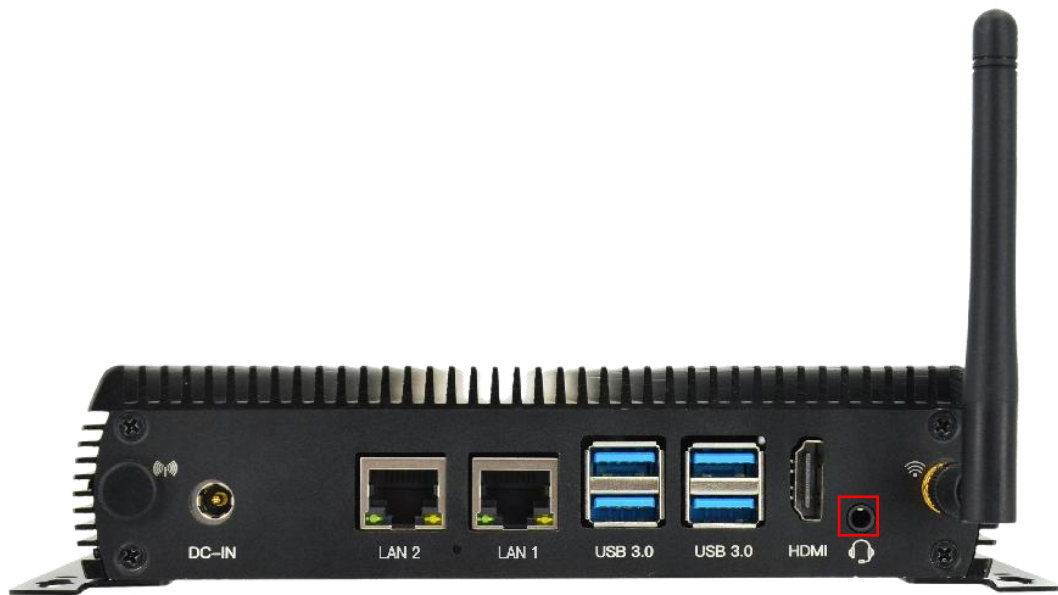


Figure 9 Audio

2.3.4. Ethernet Interface

BPC-iMX8MP-03 Industrial Computer supports two independent MAC RJ45 Gigabit Ethernet ports (Network port 1: LAN1, Network port 2: LAN2), both support POE power supply (Additional POE module is required), connect device to network through the network cable of RJ45 connector; and a set of status indicators below the interface to display the status signal, the green is Link and the yellow is Active.

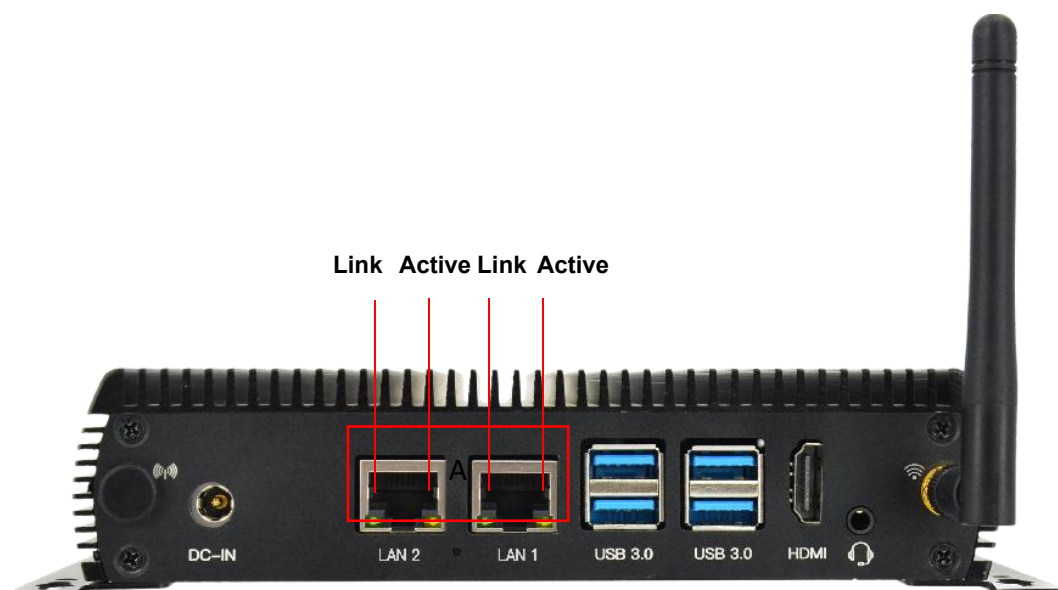


Figure 10 Ethernet

Table 3 Description of RJ45 Port Status Indicator

LED	Color	Description
Link	Green	Light, the network cable is plugged in, network connection status is good
Active	Yellow	Blinking, network data is being transmitted

2.3.5. USB Interface

BPC-iMX8MP-03 Industrial Computer has two USB controllers and PHY, supports USB 3.0 and 2.0. There are four USB 3.0 interfaces with dual-layer Type-A connectors and another two USB 2.0 interfaces with Type-A connectors. As shown in the figure below.

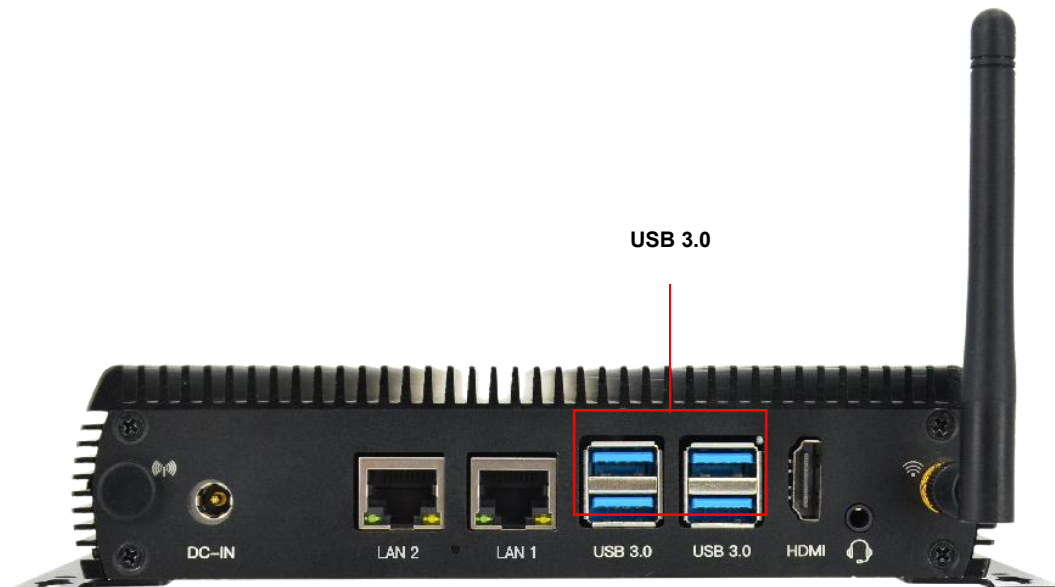


Figure 11 USB 3.0

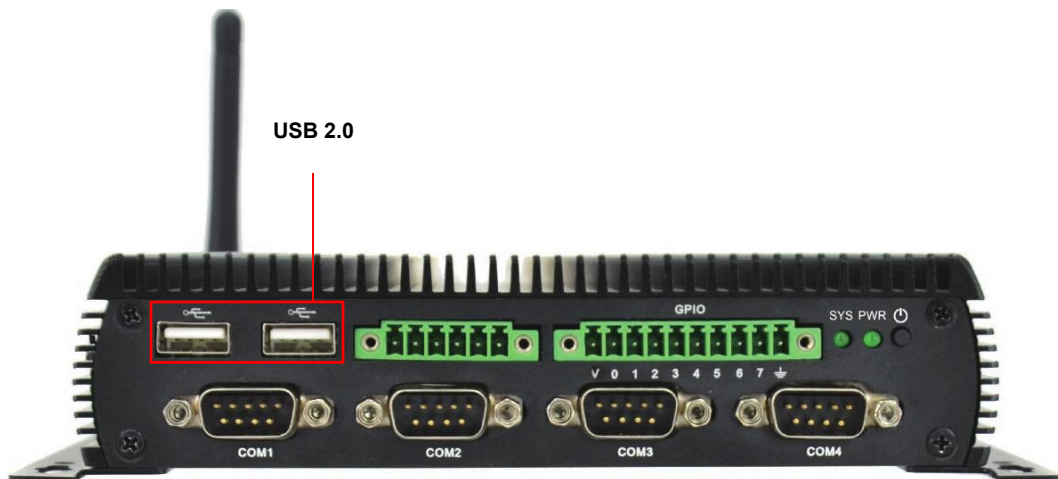


Figure 12 USB 2.0

2.3.6. RS485/RS232/CAN Interface

IMPORTANT

The default configuration is an RS485 interface. RS485, RS232, and CAN on the same connector, only one can be used at the same time. If RS232 or CAN function is required, internal hardware wiring needs to be adjusted before leaving the factory.

BPC-iMX8MP-03 Industrial Computer provides two types of serial connectors:

- Supports 2 x RS485/RS232/CAN interface, compatible with the UART TTL 3.3V serial port without physical isolation.
- Supports 4 x RS232 interfaces, compatible with UART TTL 3.3V serial ports without physical isolation.

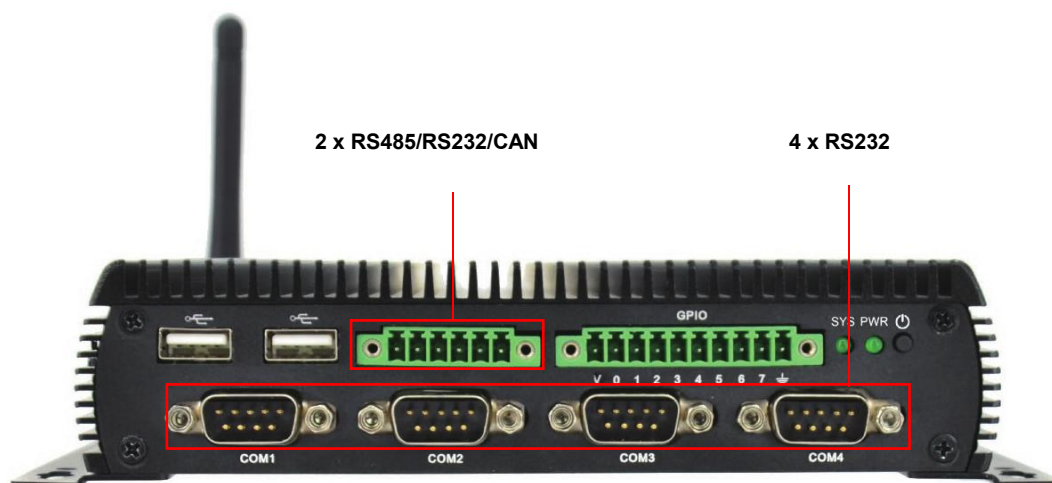


Figure 13 RS485/RS232/CAN

The 2 x RS485/RS232/CAN pin sequence is as shown in the figure:

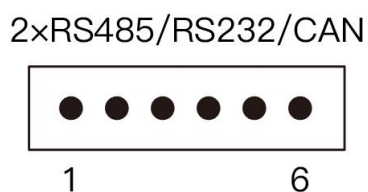


Figure 14

The 2 x RS485/RS232/CAN interface is defined as follows:

Table 4 Pin definition of RS485/RS232/CAN

Pins	Definition	Description	Device node
1	GND	RS485 differential signal line A/ RS232 sender/ CAN1 differential signal line H	/dev/ttyWCH0
2	RS485_B1/RS232_RXD1 /CAN1_L	RS485 differential signal line B/ RS232 receiver/ CAN1 differential signal line L	
3	RS485_A1/RS232_TXD1 /CAN1_H	To ground	
4	GND	To ground	/dev/ttyWCH1
5	RS485_B2/RS232_RXD2	RS485 differential signal line B/ RS232 receiver/	

	/CAN2_L	CAN2 differential signal line L	
6	RS485_A2/RS232_TXD2	RS485 differential signal line A/ RS232 sender/ CAN2	
	/CAN2_H	differential signal line H	

The 4 x RS232 pin sequence is as shown in the figure:

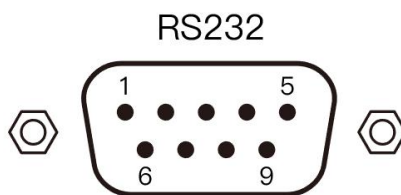


Figure 15

Table 5 Device node of COM

Function Name	IO Name	Description	Device node
COM1	RS232_RXD1	RS232 receiver	/dev/ttymxc3
	RS232_TXD1	RS232 sender	
COM2	RS232_RXD2	RS232 receiver	/dev/ttymxc2
	RS232_TXD2	RS232 sender	
COM3	RS232_RXD3	RS232 receiver	/dev/ttyWCH3
	RS232_TXD3	RS232 sender	
COM4	RS232_RXD4	RS232 receiver	/dev/ttyWCH2
	RS232_TXD4	RS232 sender	

The 4 x RS232 interface is defined as follows:

Table 6 Pin definition of COM

Pins	Definition	Pins	Definition
1	NC	2	RS232_RXD
3	RS232_TXD	4	NC
5	GND	6	NC

7	NC	8	NC
9	NC		

2.3.7. GPIO Interface

BPC-iMX8MP-03 Industrial Computer provides a GPIO interface with isolated type (from left to right, DI interface, DO interface), physically isolated DI, supports dry node input and wet node input; physically isolated DO, supports wet node, and is compatible with external relay dry nodes.

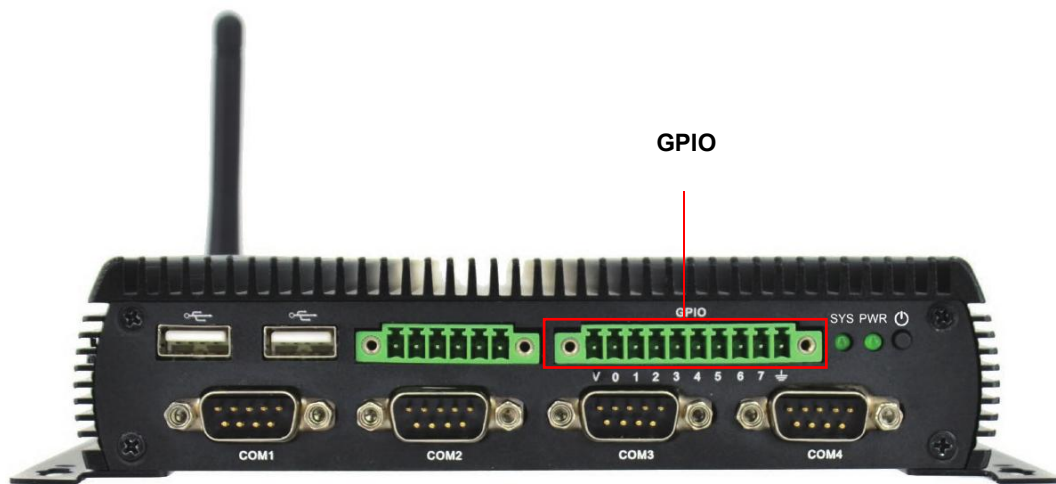


Figure 16 GPIO

The GPIO interface is defined as follows:

Table 7 Pin definition of GPIO

Pins	Definition	Pins	Definition
V	DO_PCOM	0	DI_1
1	DI_2	2	DI_3
3	DI_4	4	DO_1
5	DO_2	6	DO_3
7	DO_4	G	GND

DI electrical parameters:

- Wet contact:
 - When the signal input voltage is 0~3V DC, the corresponding SOM A signal is low level;
 - When the signal input voltage is 5~30V DC, the corresponding SOM A signal is high level.
- Dry contact:
 - Open state: high level
 - Short state with GND: low level

DO electrical parameters:

- Node DO-PCOM supports a voltage range of 5~30V DC;
- When the SOM A signal is low, the corresponding DO signal output voltage follows the node DO-PCOM, the higher the node DO-PCOM voltage, the higher the DO signal output voltage (compared with the node DO-PCOM, there is a 1~3V voltage drop);
- When the SOM A signal is high level, the corresponding DO signal output voltage is 0.

2.3.8. LED & Key

There are two LED indicators and a power ON/OFF key, as shown in the figure below.



Figure 17 LED & Key

Table 8 Description of LED & Key

LED & Key	Status	Description
SYS	Lighting	Device works normally
	off	Device works abnormally
PWR	Lighting	Power is on
	off	Power is off
ON/OFF key	Short press	Hibernation/wake up
	Long press	Power off/on

2.4. Package List

BPC-iMX8MP-03 Industrial Computer product list:

- 1 x WiFi antenna
- 4 x M3-5 Black screw
- 2 x Wall bracket
- 1 x BPC-iMX8MP-03 box

Chapter 3 Installation Guide

3.1. Installation

After receiving the product, install the accessories as follows.

1. Install the WiFi antenna to the connection port as shown in the following figure.

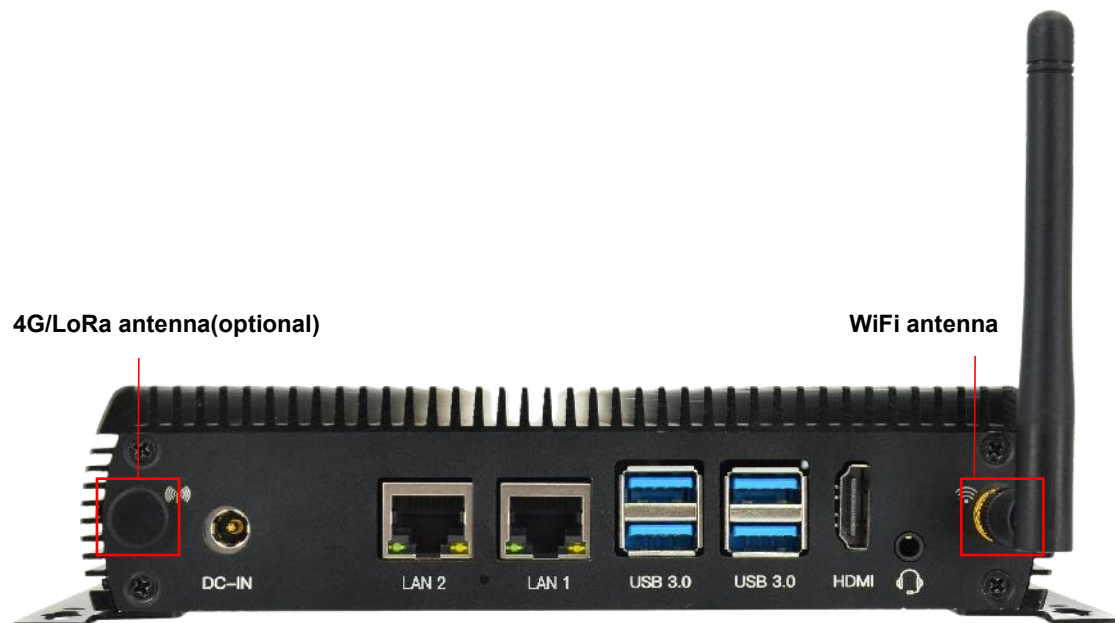


Figure 18

2. Connect the power adapter to the DC connector of enclosure (DC-IN connector as shown above). When the SYS and PWR LED are on, it proves that the Industrial Computer is powered on.



Figure 19 Power adapter

3.2. Power on

NOTE

The factory default boot mode of BPC-iMX8MP-03 Industrial Computer is eMMC boot, and **the RS232/RS485/CAN interface is configured as RS485 by default.**

If you need to change to other boot mode and the interface communication mode, please contact our engineer for modification before leaving the factory, and do not disassemble the machine by yourself.

Chapter 4 Software Application Examples

4.1. Use of Ethernet

1. Query ip command.

```
ip a
```

```
debix@imx8mpevk:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens34: <NO-CARRIER,BROADCAST,MULTICAST,DYNAMIC,UP> mtu 1500 qdisc mq state DOWN group default qlen 1000
    link/ether 10:07:23:6d:da:96 brd ff:ff:ff:ff:ff:ff
3: ens33: <BROADCAST,MULTICAST,DYNAMIC,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 10:07:23:6d:da:93 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.18/24 brd 192.168.1.255 scope global dynamic ens33
        valid_lft 78101sec preferred_lft 78101sec
    inet6 240e:36d:df5:2b00:404c:2d31:25d7:c18c/64 scope global temporary dynamic
        valid_lft 223060sec preferred_lft 77756sec
    inet6 240e:36d:df5:2b00:5d2a:4da2:a828:f1a3/64 scope global dynamic mngtmtpad
    dr noprefixroute
        valid_lft 223060sec preferred_lft 136660sec
    inet6 fe80::fd5e:df65:73dc:d698/64 scope link noprefixroute
```

As shown above: eth33 network card corresponds to the network port of the device silkscreen "LAN1" (Figure 10, right side);

eth34 network card corresponds to the network port of the device silkscreen

"LAN2" (Figure 10, left side).

2. Apply ping command.

```
ping 192.168.1.18
```

```
debix@imx8mpevk:~$ ping 192.168.1.18
PING 192.168.1.18 (192.168.1.18) 56(84) bytes of data.
64 bytes from 192.168.1.18: icmp_seq=1 ttl=64 time=0.073 ms
64 bytes from 192.168.1.18: icmp_seq=2 ttl=64 time=0.077 ms
64 bytes from 192.168.1.18: icmp_seq=3 ttl=64 time=0.081 ms
64 bytes from 192.168.1.18: icmp_seq=4 ttl=64 time=0.079 ms
64 bytes from 192.168.1.18: icmp_seq=5 ttl=64 time=0.079 ms
64 bytes from 192.168.1.18: icmp_seq=6 ttl=64 time=0.075 ms
64 bytes from 192.168.1.18: icmp_seq=7 ttl=64 time=0.071 ms
64 bytes from 192.168.1.18: icmp_seq=8 ttl=64 time=0.075 ms
64 bytes from 192.168.1.18: icmp_seq=9 ttl=64 time=0.078 ms
64 bytes from 192.168.1.18: icmp_seq=10 ttl=64 time=0.077 ms
```

3. Query the speed of the network port.

```
sudo ethtool ens33
```

```
debix@imx8mpevk:~$ sudo ethtool ens33
Settings for ens33:
    Supported ports: [ TP MII ]
    Supported link modes:   10baseT/Half 10baseT/Full
                           100baseT/Half 100baseT/Full
                           1000baseT/Full
    Supported pause frame use: Symmetric Receive-only
    Supports auto-negotiation: Yes
    Supported FEC modes: Not reported
    Advertised link modes:  10baseT/Half 10baseT/Full
                           100baseT/Half 100baseT/Full
                           1000baseT/Full
    Advertised pause frame use: Symmetric Receive-only
    Advertised auto-negotiation: Yes
    Advertised FEC modes: Not reported
    Link partner advertised link modes:  10baseT/Half 10baseT/Full
                                         100baseT/Half 100baseT/Full
                                         1000baseT/Full
    Link partner advertised pause frame use: Symmetric
    Link partner advertised auto-negotiation: Yes
    Link partner advertised FEC modes: Not reported
    Speed: 1000Mb/s
    Duplex: Full
    Port: Twisted Pair
```

The desktop settings of the BPC-iMX8MP-03 Industrial Computer Ethernet (**Settings >> Network**) are as follows.

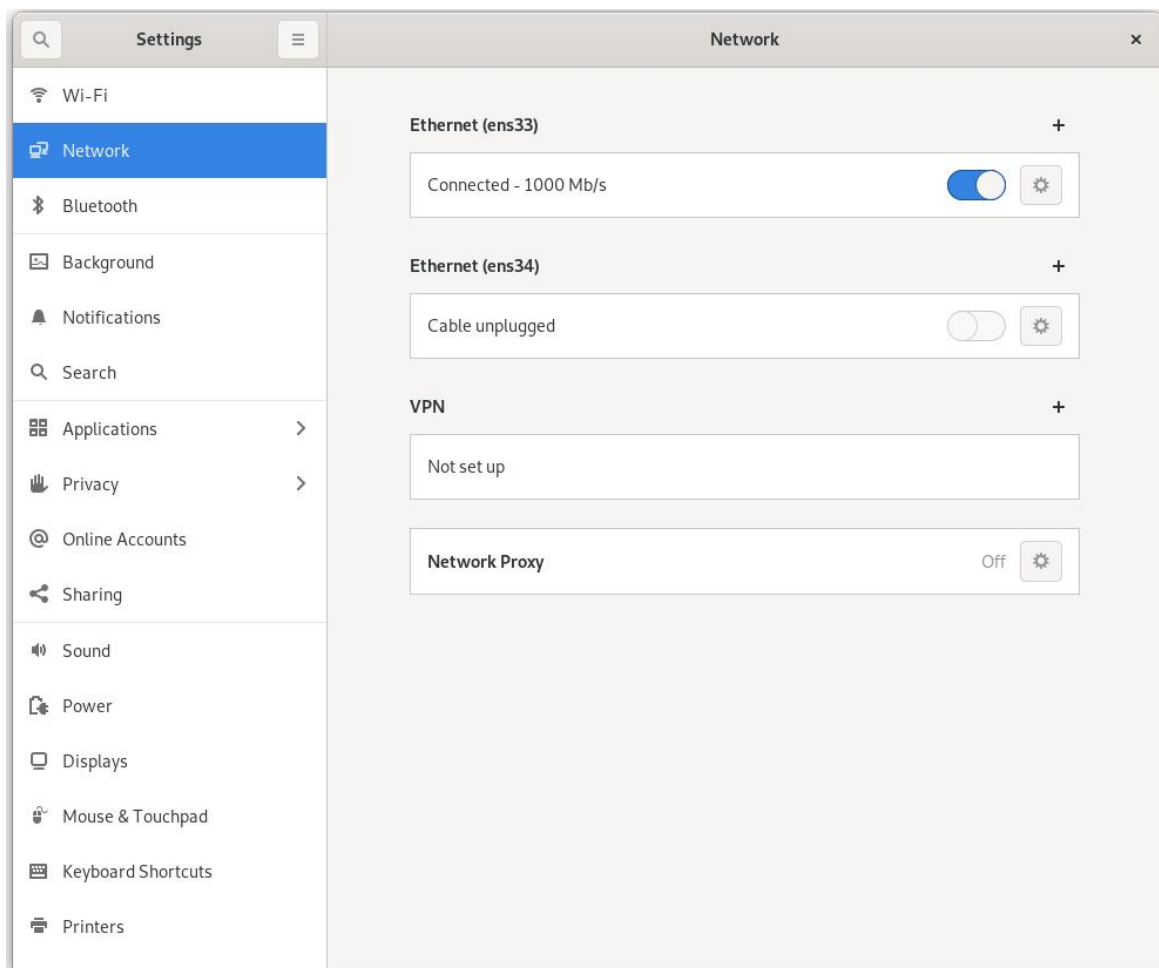


Figure 20

4.2. Use of WiFi

- Unplug the network cable, device connect WiFi (polyhex_mi), query the WiFi network port via `ifconfig wlan0`.

```
debix@imx8mpevk:~$ ifconfig wlan0
wlan0: flags=28605<UP,BROADCAST,RUNNING,MULTICAST,DYNAMIC> mtu 1500
    inet 192.168.1.21 netmask 255.255.255.0 broadcast 192.168.1.255
    inet6 fe80::48dd:f5:95c:e1a3 prefixlen 64 scopeid 0x20<link>
    inet6 240e:36d:df5:2b00:3656:9509:ddfb:e51c prefixlen 64 scopeid 0x0<global>
    inet6 240e:36d:df5:2b00:fb79:bb47:ef3f:f209 prefixlen 64 scopeid 0x0<global>
    ether ac:6a:a3:1f:b4:a7 txqueuelen 1000 (Ethernet)
    RX packets 7459 bytes 3793417 (3.7 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 8909 bytes 1261738 (1.2 MB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

- Apply ping command to check the network connection status.

```
ping 192.168.1.1
```

```
debix@imx8mpevk:~$ ping 192.168.1.1
PING 192.168.1.1 (192.168.1.1) 56(84) bytes of data.
64 bytes from 192.168.1.1: icmp_seq=1 ttl=63 time=4.65 ms
64 bytes from 192.168.1.1: icmp_seq=2 ttl=63 time=4.66 ms
64 bytes from 192.168.1.1: icmp_seq=3 ttl=63 time=4.45 ms
64 bytes from 192.168.1.1: icmp_seq=4 ttl=63 time=4.53 ms
64 bytes from 192.168.1.1: icmp_seq=5 ttl=63 time=18.6 ms
64 bytes from 192.168.1.1: icmp_seq=6 ttl=63 time=5.30 ms
64 bytes from 192.168.1.1: icmp_seq=7 ttl=63 time=4.82 ms
64 bytes from 192.168.1.1: icmp_seq=8 ttl=63 time=4.57 ms
64 bytes from 192.168.1.1: icmp_seq=9 ttl=63 time=4.47 ms
64 bytes from 192.168.1.1: icmp_seq=10 ttl=63 time=4.58 ms
64 bytes from 192.168.1.1: icmp_seq=11 ttl=63 time=5.12 ms
```

Reconnect to the new WiFi network (ChinaNet-polyhex), use the ping command, and the same screen appears as above.

The desktop settings of the BPC-IMX8MP-03 Industrial Computer WIFI (Settings >> Wi-Fi) are as follows.

- Turn on the Wi-Fi function, as shown in Figure 22.
- Click on the wifi network name, the "Authentication required" dialog box will pop up, enter the wifi network password, as shown in the figure below:



Figure 21

- Wait for a while, when you see the right side of the connected wifi name , that is, the wifi

connection is successful, as shown in the following figure.

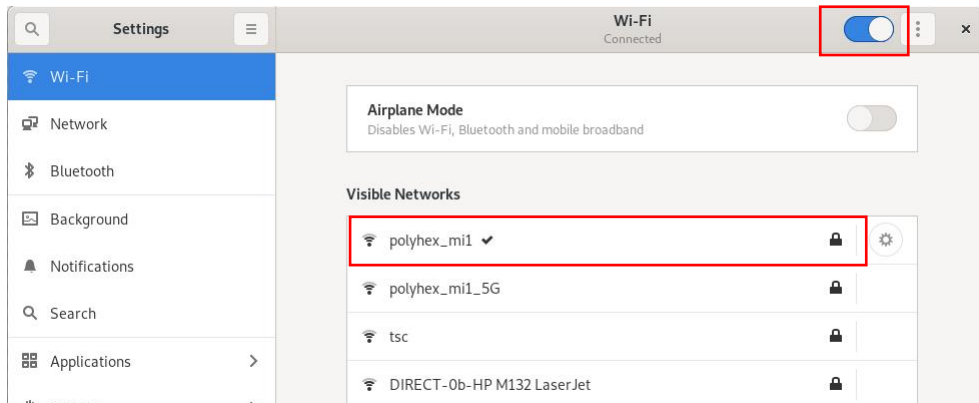


Figure 22

4.3. Use of Bluetooth

- Query Bluetooth devices via the `hciconfig` command.

```
debix@imx8mpevk:~$ hciconfig
hci0: Type: Primary Bus: UART
      BD Address: AC:6A:A3:1F:B4:A8 ACL MTU: 1021:8 SCO MTU: 64:1
      UP RUNNING PSCAN ISCAN
      RX bytes:669864 acl:167 sco:0 events:16682 errors:0
      TX bytes:11426 acl:156 sco:0 commands:464 errors:0
```

- Switch to the root user.

```
debix@imx8mpevk:~$ sudo su
root@imx8mpevk:/home/debix#
```

- Start bluetooth and match bluetooth.

```
hciconfig hci0 up
bluetoothctl
power on
agent on
default-agent
scan on
pair yourDeviceMAC      #Match the Bluetooth MAC address
```

```

root@imx8mpevk:/home/debix# hciconfig hci0 up
root@imx8mpevk:/home/debix# bluetoothctl
Agent registered
[CHG] Device 44:75:B9:BA:5E:34 RSSI: -79
[CHG] Device 18:2E:88:02:55:12 RSSI: -81
[bluetooth]# power on
[CHG] Device 00:FA:09:93:48:29 RSSI: -89
[CHG] Device 75:1B:83:44:55:38 RSSI: -85
Changing power on succeeded
[CHG] Device 00:FA:09:93:48:29 RSSI: -74
[bluetooth]# agent on
Agent is already registered
[bluetooth]# default-agent
[bluetooth]# scan on

Default agent request successful
Discovery started

```

The desktop settings of the BPC-iMX8MP-03 Industrial Computer Bluetooth (Settings >> Bluetooth) are as follows.

Example: Turn on Bluetooth on both the phone and Industrial Computer, the phone Bluetooth can detect the Bluetooth device of Industrial Computer, the Industrial Computer can detect the phone Bluetooth device, click on the Bluetooth device, connect, enter the key for pairing, as shown in the following figure.

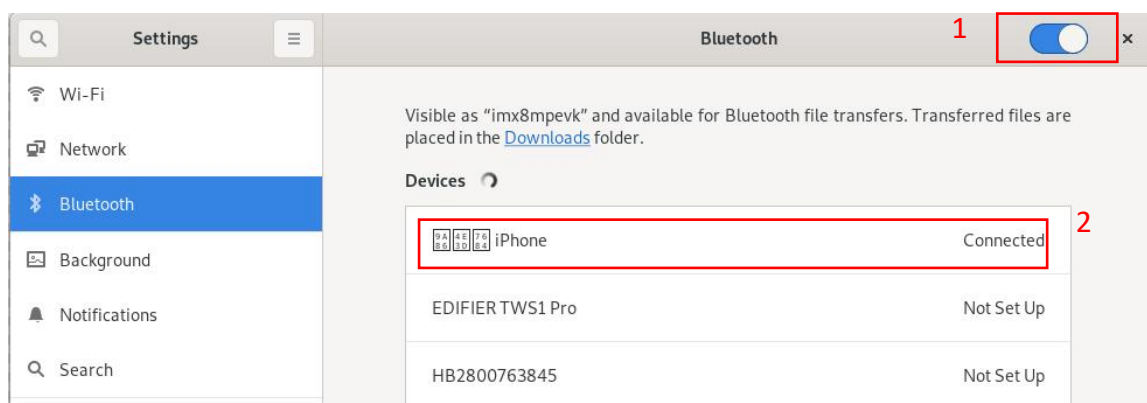


Figure 23

4.4. Use of USB

1. Switch to the root user.

```
debix@imx8mpevk:~$ sudo su
root@imx8mpevk:/home/debix#
```

2. Access the U disk in FAT32 format, the system will automatically mount it to the /mnt path.

```
df -h
```

```
root@imx8mpevk:/home/debix# df -h
Filesystem      Size  Used Avail Use% Mounted on
/dev/root        29G   3.6G   24G   14% /
devtmpfs         494M     0   494M    0% /dev
tmpfs            975M    39M   937M    4% /dev/shm
tmpfs            195M    2.1M   193M    2% /run
tmpfs            5.0M    4.0K    5.0M    1% /run/lock
tmpfs            975M     0   975M    0% /sys/fs/cgroup
/dev/mmcblk1p1   500M    31M   470M    7% /boot
tmpfs            195M    44K   195M    1% /run/user/1000
/dev/sda1        253M    31M   222M   12% /media/debix/boot
/dev/sda2        15G    3.6G    11G   26% /media/debix/rootfs
```

- If the U disk is not mounted, you can mount the U disk with the following command:

- Query the U disk letter:

```
fdisk -l
```

```
root@imx8mpevk:/home/debix# fdisk -l
Disk /dev/mtdblock0: 8 MiB, 8388608 bytes, 16384 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/mmcblk2: 14.57 GiB, 15634268160 bytes, 30535680 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xfc26a6dc

Device         Boot    Start        End    Sectors    Size Id Type
/dev/mmcblk2p1      20480    1024000    1003521    490M 83 Linux
/dev/mmcblk2p2   1228800   30535679   29306880    14G 83 Linux

Disk /dev/mmcblk1: 29.74 GiB, 31914983424 bytes, 62333952 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
```

```
Disk identifier: 0x00dba0b

Device      Boot   Start     End  Sectors  Size Id Type
/dev/mmcblk1p1      20480  1044479  1024000   500M  c W95 FAT32 (LBA)
/dev/mmcblk1p2    1228800  62333951  61105152  29.1G  83 Linux

Disk /dev/sda: 14.86 GiB, 15931539456 bytes, 31116288 sectors
Disk model: STORAGE DEVICE
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x994d7a7d

Device      Boot   Start     End  Sectors  Size Id Type
/dev/sda1      8192     532479   524288   256M  c W95 FAT32 (LBA)
/dev/sda2    532480  31116287  30583808  14.6G  83 Linux
```

- Mounting the U disk:

```
mount /dev/sda1 /mnt
```

3. Enter the U disk directory:

```
cd /mnt
```

```
root@imx8mpevk:/home/debix# cd /mnt
root@imx8mpevk:/mnt# ls
COPYING.linux          bcm2711-rpi-cm4s.dtb  issue.txt
LICENCE.broadcom       bootcode.bin         kernel8.img
'System Volume Information' cmdline.txt           overlays
bcm2710-rpi-2-b.dtb    config.txt            start.elf
bcm2710-rpi-3-b-plus.dtb fixup.dat             start4.elf
bcm2710-rpi-3-b.dtb    fixup4.dat           start4cd.elf
bcm2710-rpi-cm3.dtb    fixup4cd.dat         start4db.elf
bcm2710-rpi-zero-2-w.dtb fixup4db.dat         start4x.elf
bcm2710-rpi-zero-2.dtb fixup4x.dat          start_cd.elf
bcm2711-rpi-4-b.dtb    fixup_cd.dat         start_db.elf
bcm2711-rpi-400.dtb    fixup_db.dat         start_x.elf
bcm2711-rpi-cm4.dtb    fixup_x.dat
```

4. Clear the cache, run before each read and write test command.

```
sh -c "sync && echo 3 > /proc/sys/vm/drop_caches"
```

```
root@imx8mpevk:/home/debix# mount /dev/sda1 /mnt/
root@imx8mpevk:/home/debix# cd /mnt/
root@imx8mpevk:/mnt# sh -c "sync && echo 3 > /proc/sys/vm/drop_caches"
```

5. Write speed test.

```
sh -c "sync && echo 3 > /proc/sys/vm/drop_caches" // clear cache

dd if=/dev/zero of=./test_write count=1 bs=1G
```



```
root@imx8mpevk:/mnt# dd if=/dev/zero of=./test_write count=1 bs=1G
1+0 records in
1+0 records out
1073741824 bytes (1.1 GB, 1.0 GiB) copied, 26.6288 s, 40.3 MB/s
```

6. Reading speed test.

```
sh -c "sync && echo 3 > /proc/sys/vm/drop_caches" // clear cache

dd if=./test_write of=/dev/null count=1 bs=1G
```

```
root@imx8mpevk:/mnt# sh -c "sync && echo 3 > /proc/sys/vm/drop_caches"
root@imx8mpevk:/mnt# dd if=./test_write of=/dev/null count=1 bs=1G
1+0 records in
1+0 records out
1073741824 bytes (1.1 GB, 1.0 GiB) copied, 43.7707 s, 24.5 MB/s
```

4.5. Verification of RS485/RS232/CAN

IMPORTANT

The default configuration is an RS485 interface. RS485, RS232, and CAN on the same connector, only one can be used at the same time. If RS232 or CAN function is required, internal hardware wiring needs to be adjusted before leaving the factory.

1. Install cutecom serial port tool on BPC-iMX8MP-03 Industrial Computer.

```
sudo apt update

sudo apt install cutecom qtwayland5
```

2. The serial port parameters are set as follows.

Table 9 Cutecom parameter settings

Parameter	Value
Baudrate	115200
Data Bits	8
Stop Bits	1
Parity	None
Flow Control	None

4.5.1. RS485 (Default)

Connect Pin2 to Pin5, and connect Pin3 to Pin6 (that is, A to A, and B to B). The wiring is shown in the following figure.

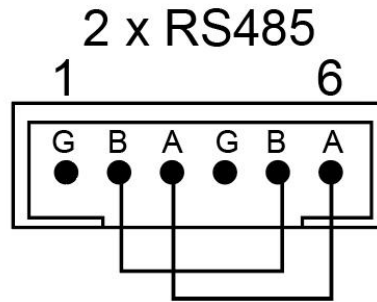


Figure 24

- Run the `cutecom` tool, set the **Device** to `/dev/ttyWCH0`, set other parameters as shown in the table, click **Open**.

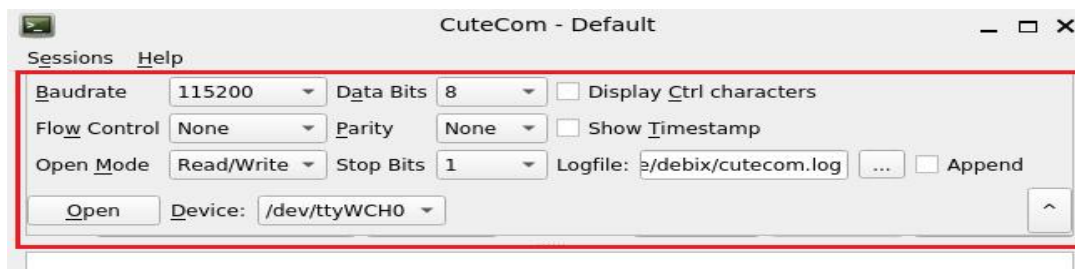


Figure 25

- Open another window of the `cutecom` tool, set the **Device** to `/dev/ttyWCH1` and click **Open**.
- Send and receive data via `cutecom`. Enter the test string in the `cutecom` input box, press **Enter** to send, you can see that another `cutecom` receiving box receives the same message, indicating that the communication is successful, and the result is as follows:

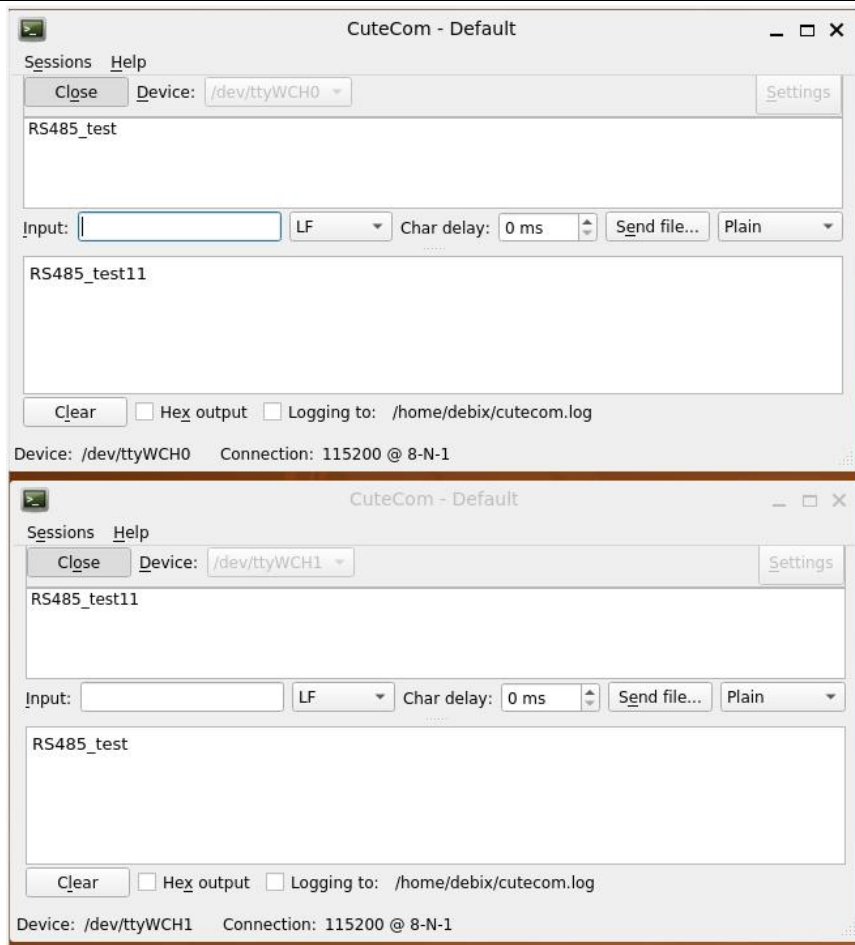


Figure 26

4.5.2. RS232

Connect Pin2 of RS232 to the sending end of USB-RS232, Pin3 to the receiving end of USB-RS232, and Pin1 to the ground terminal of USB-RS232, USB-RS232 is connected to the onboard USB 3.0 interface. The wiring is shown in the following figure.

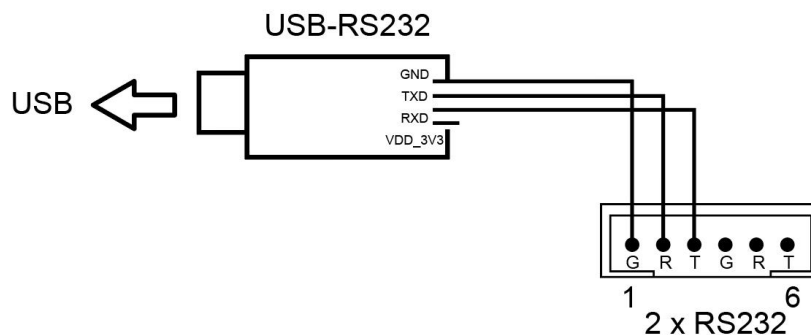


Figure 27

- Set the **Device** of the **cutecom** tool to **/dev/ttyWCH0**, click **Open**.
- Set the **Device** of another **cutecom** tool to **/dev/ttyUSB0**, set other parameters as below, click **Open**.

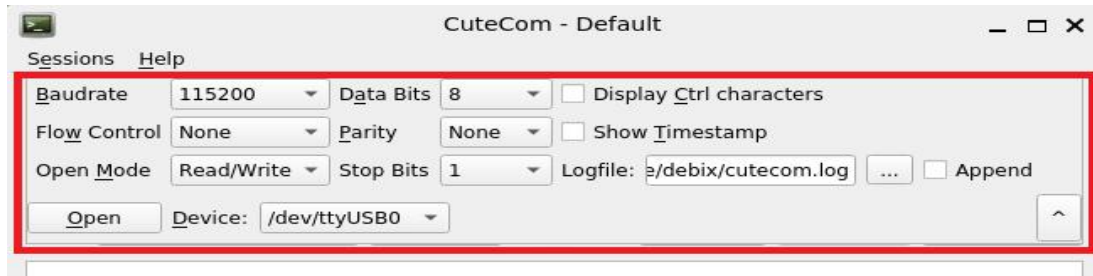


Figure 28

- Send and receive data via cutecom. Enter the test string in the cutecom input box, press **Enter** to send, you can see that another cutecom receiving box receives the same message, indicating that the communication is successful, and the result is as follows:

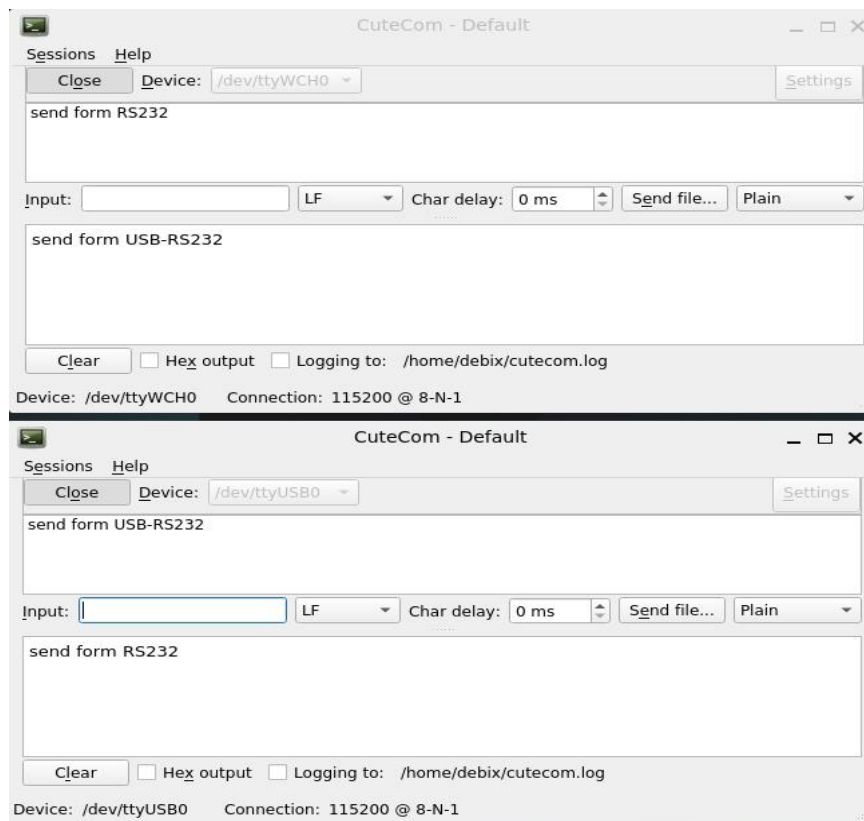


Figure 29

4.5.3. CAN

Connect Pin2 to Pin5, and connect Pin3 to Pin6 (that is, H to H, and L to L). The wiring is shown in the following figure.

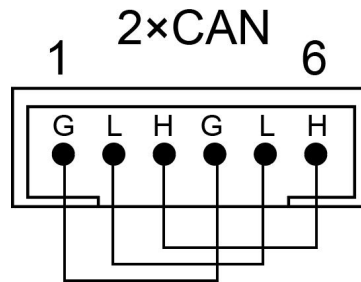


Figure 30

Switch to the root user first.

```
sudo su
```

```
debix@imx8mpevk:~$ sudo su
root@imx8mpevk:/home/debix#
```

- CAN1 sends data, CAN0 receives data

1. Open a Terminal and configure CAN0 to receive.

```
ifconfig can0 down
ip link set can0 type can bitrate 500000
ifconfig can0 up
candump can0
```

```
root@imx8mpevk: /home/debix
debix@imx8mpevk:~$ sudo su
[sudo] password for debix:
root@imx8mpevk:/home/debix#
root@imx8mpevk:/home/debix# ifconfig can0 down
root@imx8mpevk:/home/debix# ip link set can0 type can bitrate 500000
root@imx8mpevk:/home/debix# ifconfig can0 up
root@imx8mpevk:/home/debix# candump can0
can0 123 [8] 11 22 33 44 55 66 77 88
```

2. Open another Terminal and configure CAN1 to send.

```
ifconfig can1 down
ip link set can1 type can bitrate 500000
ifconfig can1 up
cansend can1 123#1122334455667788
```



```
root@imx8mpevk: /home/debix
debix@imx8mpevk:~$ sudo su
[sudo] password for debix:
root@imx8mpevk:/home/debix# ifconfig can1 down
root@imx8mpevk:/home/debix# ip link set can1 type can bitrate 500000
root@imx8mpevk:/home/debix# ifconfig can1 up
root@imx8mpevk:/home/debix# cansend can1 123#1122334455667788
```

- CAN1 receives data, CAN0 sends data

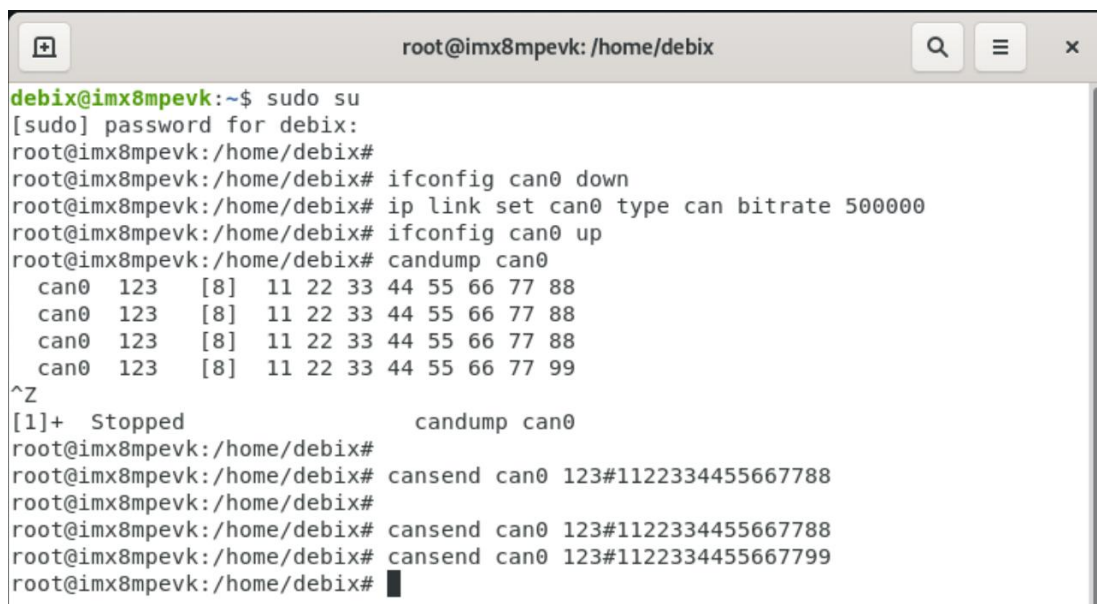
1. In the Terminal sent by CAN1, switch can1 to receiving state.

```
candump can1
```

2. In the Terminal receiving CAN0, press **Ctrl+Z** to end receiving. And switch can0 to send state.

```
cansend can0 123#1122334455667788
```

As shown in the figure below:



```
root@imx8mpevk: /home/debix
debix@imx8mpevk:~$ sudo su
[sudo] password for debix:
root@imx8mpevk:/home/debix# ifconfig can0 down
root@imx8mpevk:/home/debix# ip link set can0 type can bitrate 500000
root@imx8mpevk:/home/debix# ifconfig can0 up
root@imx8mpevk:/home/debix# candump can0
can0 123 [8] 11 22 33 44 55 66 77 88
can0 123 [8] 11 22 33 44 55 66 77 88
can0 123 [8] 11 22 33 44 55 66 77 88
can0 123 [8] 11 22 33 44 55 66 77 99
^Z
[1]+  Stopped                  candump can0
root@imx8mpevk:/home/debix#
root@imx8mpevk:/home/debix# cansend can0 123#1122334455667788
root@imx8mpevk:/home/debix#
root@imx8mpevk:/home/debix# cansend can0 123#1122334455667788
root@imx8mpevk:/home/debix# cansend can0 123#1122334455667799
root@imx8mpevk:/home/debix#
```

```
root@imx8mpevk: /home/debix
debix@imx8mpevk:~$ sudo su
[sudo] password for debix:
root@imx8mpevk:/home/debix# ifconfig can1 down
root@imx8mpevk:/home/debix# ip link set can1 type can bitrate 500000
root@imx8mpevk:/home/debix# ifconfig can1 up
root@imx8mpevk:/home/debix# cansend can1 123#1122334455667788
root@imx8mpevk:/home/debix# cansend can1 123#1122334455667788
root@imx8mpevk:/home/debix# cansend can1 123#1122334455667788
root@imx8mpevk:/home/debix# cansend can1 123#1122334455667799
root@imx8mpevk:/home/debix# candump can1
can1 123 [8] 11 22 33 44 55 66 77 88
can1 123 [8] 11 22 33 44 55 66 77 88
can1 123 [8] 11 22 33 44 55 66 77 99
```

4.6. Verification of DI/DO

Switch to the root user first.

```
debix@imx8mpevk:~$ sudo su
root@imx8mpevk:/home/debix#
```

4.6.1. DI

Take the dry node connection as an example, all ports of DI operate in the same way, here take DI_1 as an example, and connect the DI_1 to ground in series.

1. Enter the GPIO directory.

```
cd /sys/class/gpio
```

```
root@imx8mpevk:~# cd /sys/class/gpio
root@imx8mpevk:/sys/class/gpio# ls
export      gpiochip128  gpiochip480  gpiochip64  unexport
gpiochip0   gpiochip32   gpiochip496  gpiochip96
```

2. Export GPIO. After the export is successful, the gpio5 directory will be automatically generated, as shown below:

```
echo 5 > export
```

```
root@imx8mpevk:/sys/class/gpio# echo 5 > export
root@imx8mpevk:/sys/class/gpio# ls
export  gpiochip0   gpiochip32  gpiochip496  gpiochip96
gpio5   gpiochip128  gpiochip480  gpiochip64   unexport
root@imx8mpevk:/sys/class/gpio#
```


3. Configure GPIO pins.

```
echo in > gpio5/direction //Set pin direction to input
echo none > gpio5/edge //set non-interrupt pin
```

```
root@mx8mpevk:/sys/class/gpio# echo in > gpio5/direction
root@mx8mpevk:/sys/class/gpio# echo none > gpio5/edge
root@mx8mpevk:/sys/class/gpio# cat gpio5/value
1
root@mx8mpevk:/sys/class/gpio#
```

4.6.2. DO

The current supported by DO port is 500mA, take DO_1 as an example.

1. Enter the DOUT_1 control directory

```
cd /sys/devices/platform/gpio-leds/leds/DOUT_1
```

2. GPIO output is low, DO_1 open-drain output is high configuration, and the relay is turned off.

```
echo 0 > brightness
```

3. GPIO output is high, DO_1 output is low, and the relay is on.

```
echo 1 > brightness
```

The other DO verification methods are the same, the DOUT_1, DOUT_2, DOUT_3, and DOUT_4 files under /sys/devices/platform/gpio-leds/leds/ correspond to DO_1, DO_2, DO_3, and DO_4 ports respectively.

```
root@mx8mpevk:~# cd /sys/devices/platform/gpio-leds/leds/
root@mx8mpevk:/sys/devices/platform/gpio-leds/leds# ls
4G_RESET      DOUT_1        GPIO_LED2     SPDIF_TX      VDD5V_EN
BB_VDD5V_EN    DOUT_2        S0M_VDD1V8_EN USB20_PWR_EN   yellow:status
BB_VDD5V_EN_1  DOUT_3        S0M_VDD3V3_EN USB30_OTG_EN
CSI1_VDD1V8_EN DOUT_4        SPDIF_EXT_CLK USB30_PWR_EN
CSI1_VDD3V3_EN GPIO_LED1     SPDIF_RX      USB30_RST
root@mx8mpevk:/sys/devices/platform/gpio-leds/leds#
```

4.7. Verification of LED & Key

1. LED

- SYS is the system status indicator, the indicator is on when the Industrial Computer is running normally; otherwise, the indicator is off.
- PWR is the power status indicator, the indicator is on when power is applied; otherwise, the indicator is off when power is off.

2. Key

BPC-iMX8MP-03 Industrial Computer is automatically turned on when power is applied.

- Short press
 - SYS green light is off, the system enters into sleep.
 - Short press again, SYS green light is on to wake up the system.
- Long press
 - Press and hold until the green light turns off to shut down.
 - Press and hold again until the green light turns on to boot the system.

4.8. Verification of RTC

Confirm that the HYM8653S driver module is loaded successfully.

```
dmesg | grep rtc-hym8653
```

The desktop settings of the BPC-iMX8MP-03 Industrial Computer RTC (**Settings >> Date&Time**) are as follows.

- Unlock "Date & Time" to turn on or off automatic time.
- Set "Time Zone" as local zone.
- Set "Time Format" to 24-hour.

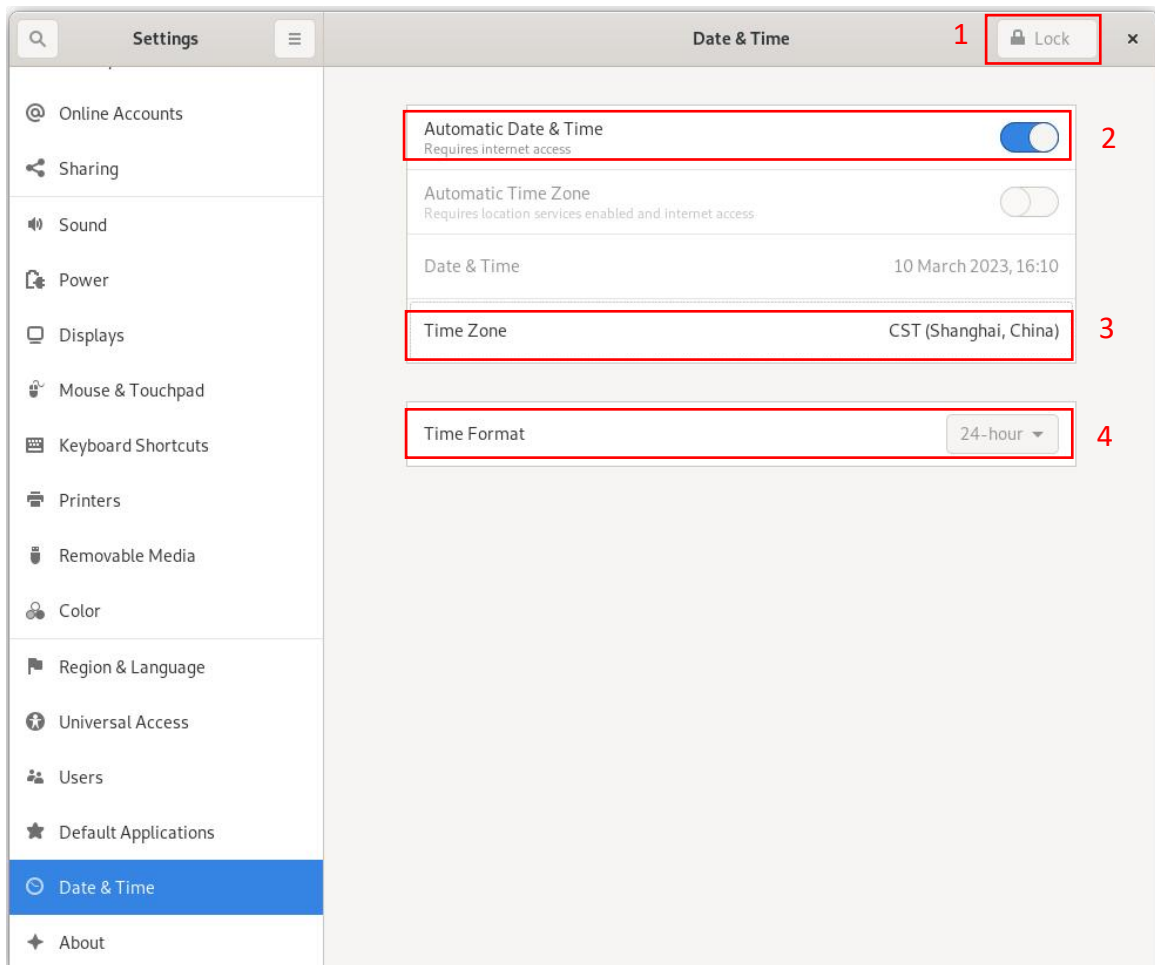


Figure 31

Read the RTC time of the Industrial Computer via `hwclock -r` command, as shown in the following figure.

```
root@mx8mpevk:~# hwclock -r
2023-03-10 16:21:44.971233+08:00
root@mx8mpevk:~#
```